

Simulation Study Plan

Terry Xue

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1. Introduction

This document presents a simulation study plan for evaluating a lognormal extension of the Vink et al. transmission-path mixture framework for serial interval (SI) estimation. The plan follows the ADEMP structure: Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.

The serial interval is defined as the time between symptom onset in a primary case and symptom onset in the corresponding secondary case. It is an important epidemiological quantity because it describes the timing of transmission and is used in epidemic models, reproduction number estimation, and public health planning. In most outbreak datasets, however, true infector–infectee pairs are not observed. We usually observe symptom-onset dates only. Therefore, the SI must be inferred indirectly.

Following Vink et al. (2014), the observable quantity is the index case-to-case (ICC) interval: the time from symptom onset in the index case of a cluster to symptom onset in each later case. An ICC interval is not necessarily a direct serial interval. It may arise from several latent transmission paths: co-primary (CP), primary-secondary (PS), primary-tertiary (PT), or primary-quaternary (PQ) transmission. The PS path is the target component, because its ICC interval is exactly one direct SI. The other paths are nuisance components that must be represented to recover the PS distribution from the observed ICC mixture.

The current project extends the original Vink framework by replacing the Normal or Gamma assumption for the underlying SI distribution with a lognormal assumption. In this updated plan, the extension retains the original transmission-path interpretation: all direct serial intervals are assumed to be independent and identically distributed copies from one underlying SI distribution. This is different from an earlier structurally constrained lognormal mixture plan, where CP, PS, PT, and PQ were directly specified as four ordered lognormal components with fixed path multipliers. That earlier formulation is not used here.

The proposed simulation study has two main purposes. First, under correct specification, it evaluates whether the iid lognormal extension can recover the true natural-scale SI mean and SD. Second, under misspecification, it evaluates how robust the lognormal extension is when the true underlying SI distribution is Gamma or Normal rather than lognormal. Formal model comparison against Normal and Gamma fitted models is not included in this first simulation plan.

2. Simulation Study

2.1 Aims

2.1.1 Primary aim

To assess whether the proposed iid lognormal extension of the Vink transmission-path mixture framework can recover the mean and SD of the underlying SI distribution when the data are generated under the same iid lognormal mechanism assumed by the fitted model.

2.1.2 Secondary aims

- To examine the performance of the proposed method under correct specification.
- To examine sensitivity to different latent transmission-path mixture weights under correct specification.
- To examine robustness when the fitted model assumes an iid lognormal SI but the true SI distribution is Gamma or Normal.

2.1.3 Questions of interest

The simulation study is designed to answer the following questions:

- Under correct specification, can the lognormal extension recover the true SI mean and SD properly?
- How does performance change as the number of observed ICC intervals increases?
- How sensitive is estimation to the relative weights of CP, PS, PT, and PQ paths in the observed ICC data?
- If the true SI distribution is Gamma or Normal, how biased are the SI mean and SD estimates from the lognormal extension?

2.2 General data-generating mechanism

For each simulated dataset, let the latent transmission path for observation i be

$$Z_i \in \{CP, PS, PT, PQ\},$$

with path probabilities

$$\mathbf{w} = (w_{CP}, w_{PS}, w_{PT}, w_{PQ}), \quad \sum_g w_g = 1.$$

The underlying direct serial intervals are independent random variables drawn from a specified true SI distribution. Under correct specification, this true distribution is lognormal:

$$X_k \stackrel{i.i.d.}{\sim} \text{Lognormal}(\text{meanlog}, \text{sdlog}^2), \quad k = 1, 2, 3, \dots$$

where meanlog and sdlog are log-scale parameters. The natural-scale SI mean and SD are denoted by μ and σ :

$$\mu = E(X) = \exp\left(\text{meanlog} + \frac{1}{2}\text{sdlog}^2\right),$$

and

$$\sigma = SD(X) = \sqrt{\left[\exp(\text{sdlog}^2) - 1\right] \exp(2\text{meanlog} + \text{sdlog}^2)}.$$

The path-specific continuous ICC interval C_i is generated as follows:

$$C_{CP} = |X_1 - X_2|,$$

$$C_{PS} = X_1,$$

$$C_{PT} = X_1 + X_2,$$

and

$$C_{PQ} = X_1 + X_2 + X_3.$$

This preserves the original Vink transmission-path interpretation. The primary-secondary path is exactly the target SI distribution, while CP, PT, and PQ are derived from the same underlying iid SI distribution.

3. Scenario Design

The simulation study contains two scenario groups:

1. Correct specification scenarios, where the true SI distribution is lognormal and the fitted model is also lognormal.
2. Misspecification scenarios, where the fitted model remains lognormal but the true SI distribution is Gamma or Normal.

Unless otherwise stated, the baseline natural-scale SI parameters are

$$\mu = 6 \text{ days}, \quad \sigma = 3 \text{ days}.$$

For lognormal data generation, these natural-scale values are converted to log-scale parameters by

$$\text{sdlog} = \sqrt{\log \left(1 + \frac{\sigma^2}{\mu^2} \right)},$$

and

$$\text{meanlog} = \log(\mu) - \frac{1}{2}\text{sdlog}^2.$$

3.1 Scenario A: Correct specification

In all correct-specification scenarios, data are generated from the same iid lognormal SI mechanism assumed by the fitted model:

$$X \sim \text{Lognormal}(\text{meanlog}, \text{sdlog}^2).$$

The fitted method is the proposed lognormal extension implemented through `mitey::si_estim(dat, dist = "lognormal")`.

A0. Baseline scenario

The baseline scenario checks whether the method can recover the true SI mean and SD in a standard setting.

Parameter	Value
True SI distribution	Lognormal
True SI mean	6 days
True SI SD	3 days
ICC sample size	300
Path weights	(0.20, 0.50, 0.20, 0.10) for CP, PS, PT, PQ
Number of simulation replicates	500
Fitted model	Lognormal extension

A1. Finite sample size

This scenario evaluates finite-sample performance. Only the number of observed ICC intervals changes; all other baseline parameters are fixed.

$$n_{obs} \in \{100, 300, 600, 1000\}.$$

The purpose is to assess whether estimation becomes less biased and more precise as the number of observed ICC intervals increases.

Scenario	n_{obs}	True SI distribution	True SI mean	True SI SD	Path weights
A1-n100	100	Lognormal	6	3	(0.20, 0.50, 0.20, 0.10)
A1-n300	300	Lognormal	6	3	(0.20, 0.50, 0.20, 0.10)
A1-n600	600	Lognormal	6	3	(0.20, 0.50, 0.20, 0.10)
A1-n1000	1000	Lognormal	6	3	(0.20, 0.50, 0.20, 0.10)

A2. Mixture weight patterns

This scenario evaluates sensitivity to the relative proportions of latent transmission paths. The true SI distribution and sample size are fixed at the baseline values. Only the path weights change.

Scenario	CP	PS	PT	PQ	Purpose
A2-baseline	0.20	0.50	0.20	0.10	Standard mixture composition
A2-PS-dominant	0.10	0.70	0.15	0.05	Strong PS signal
A2-CP-heavy	0.40	0.35	0.15	0.10	More co-primary intervals
A2-higher-generation-heavy	0.10	0.35	0.35	0.20	More PT/PQ intervals

The main question is whether the PS/SI mean and SD remain recoverable when the PS component is less dominant or when nuisance components make up a larger proportion of the observed ICC intervals.

3.2 Scenario B: Misspecification

In misspecification scenarios, data are generated from a true SI distribution that is not lognormal. The fitted model is still the proposed lognormal extension:

```
mitey::si_estim(dat, dist = "lognormal")
```

, which aims to evaluate robustness of the lognormal extension under misspecified distribution setting for SI.

B1. True SI is Gamma

In this scenario, the true direct SI distribution is Gamma with the same natural-scale mean and SD as the baseline scenario:

$$X \sim \text{Gamma}(\mu = 6, \sigma = 3).$$

In implementation, the Gamma parameters are

$$shape = \frac{\mu^2}{\sigma^2}, \quad scale = \frac{\sigma^2}{\mu}.$$

The ICC intervals are generated using the same path construction:

$$CP = |X_1 - X_2|, \quad PS = X_1, \quad PT = X_1 + X_2, \quad PQ = X_1 + X_2 + X_3.$$

Parameter	Value
True SI distribution	Gamma
True SI mean	6 days
True SI SD	3 days
ICC sample size	300
Path weights	(0.20, 0.50, 0.20, 0.10)
Number of simulation replicates	500
Fitted model	Lognormal extension

B2. True SI is Normal

In this scenario, the true direct SI distribution follows the Normal assumption.

The true direct SI distribution is

$$X \sim \text{Normal}(6, 3^2).$$

Observed ICC intervals are generated using a folded Vink-style construction so that the generated ICC intervals remain non-negative:

$$CP = |X_1 - X_2|,$$

$$PS = |X_1|,$$

$$PT = |X_1 + X_2|,$$

and

$$PQ = |X_1 + X_2 + X_3|.$$

The fitted model remains the lognormal extension. The main question is whether the lognormal extension produces substantial bias when the data are generated from the Normal assumption.

Parameter	Value
True SI distribution	Normal
True SI mean	6 days
True SI SD	3 days
ICC sample size	300
Path weights	(0.20, 0.50, 0.20, 0.10)
Number of simulation replicates	500
Fitted model	Lognormal extension

4. Estimands and Targets

The primary estimands are the natural-scale mean and SD of the underlying direct SI distribution:

$$\theta_1 = \mu = E(X), \quad \theta_2 = \sigma = SD(X).$$

These are the main quantities reported by `si_estim()` and are the most important epidemiological summaries for modelling.

For correct-specification scenarios, the true values are those of the underlying lognormal SI distribution. For misspecification scenarios, the true values are the natural-scale mean and SD of the data-generating SI distribution, such as Gamma or Normal.

Secondary quantities to record are:

- estimated mixture weights;
- convergence indicator;
- number of EM iterations used;

Since these secondary quantities are diagnostic, the main performance conclusions should be based on recovery of the SI mean and SD.

5. Methods

5.1 Fitted method

The fitted method in all scenarios is the proposed lognormal extension of the Vink framework. The code is

```
fit <- mitey::si_estim(  
  dat = observed_icc_intervals,  
  dist = "lognormal",  
  n = 50,  
  n_starts = 1  
)
```

The fitted model treats each observed ICC interval as arising from one of four latent paths: CP, PS, PT, or PQ. It estimates the SI mean and SD, along with mixture weights for the path components.

The E-step calculates posterior path responsibilities using the current parameter estimates and the path-specific probabilities. Under the lognormal extension, CP, PT, and PQ involve differences or sums of lognormal random variables and therefore require numerical integration. The M-step updates the path weights and the lognormal SI parameters.

5.2 Simulation execution using mitey package

For SI estimation, we will generate CP, PS, PT, and PQ components, combines them into an ICC interval vector, and then fits `si_estim()`.

The code is

```
set.seed(1234)  
  
N <- 300  
mu <- 6  
sigma <- 3  
route_weights <- c(CP = 0.20, PS = 0.50, PT = 0.20, PQ = 0.10)  
  
sdlog <- sqrt(log(1 + sigma^2 / mu^2))  
meanlog <- log(mu) - 0.5 * sdlog^2  
  
n_route <- setNames(as.integer(round(route_weights * N)), names(route_weights))  
n_route["PS"] <- N - sum(n_route[c("CP", "PT", "PQ")])  
  
CP <- abs(  
  rlnorm(n_route["CP"], meanlog = meanlog, sdlog = sdlog) -  
  rlnorm(n_route["CP"], meanlog = meanlog, sdlog = sdlog)  
)  
PS <- rlnorm(n_route["PS"], meanlog = meanlog, sdlog = sdlog)  
PT <- rlnorm(n_route["PT"], meanlog = meanlog, sdlog = sdlog) +  
  rlnorm(n_route["PT"], meanlog = meanlog, sdlog = sdlog)  
PQ <- rlnorm(n_route["PQ"], meanlog = meanlog, sdlog = sdlog) +  
  rlnorm(n_route["PQ"], meanlog = meanlog, sdlog = sdlog) +
```



```

  rlnorm(n_route["PQ"], meanlog = meanlog, sdlog = sdlog)

observed_icc_intervals <- round(c(CP, PS, PT, PQ))
observed_icc_intervals <- pmax(observed_icc_intervals, 0)

fit <- mitey::si_estim(
  dat = observed_icc_intervals,
  dist = "lognormal",
  n = 50,
  n_starts = 1
)

```

The output for each replicate should contain:

```

data.frame(
  scenario_id = scenario_id,
  replicate_id = replicate_id,
  n_obs = n_obs,
  true_dist = si_dist,
  fit_dist = fit_dist,
  true_mean = mu,
  true_sd = sigma,
  est_mean = fit$mean,
  est_sd = fit$sd,
  converged = fit$converged,
  iterations = fit$iterations
)

```

The simulation scenarios should be controlled by a scenario table:

```

scenario_grid <- data.frame(
  scenario_id = c(
    "A0_baseline",
    "A1_n100", "A1_n300", "A1_n600", "A1_n1000",
    "A2_baseline", "A2_PS_dominant", "A2_CP_heavy",
    "A2_higher_generation_heavy",
    "B1_gamma", "B2_normal"
  ),
  scenario_type = c(
    "correct",
    rep("correct", 8),
    "misspecified", "misspecified"
  ),
  si_dist = c(
    "lognormal",
    rep("lognormal", 8),
    "gamma", "normal"
  ),
  n_obs = c(

```

```

    300,
    100, 300, 600, 1000,
    300, 300, 300, 300,
    300, 300
  ),
  true_mean = 6,
  true_sd = 3,
  w_cp = c(0.20, 0.20, 0.20, 0.20, 0.20, 0.20, 0.10, 0.40, 0.10, 0.20, 0.20),
  w_ps = c(0.50, 0.50, 0.50, 0.50, 0.50, 0.50, 0.70, 0.35, 0.35, 0.50, 0.50),
  w_pt = c(0.20, 0.20, 0.20, 0.20, 0.20, 0.20, 0.15, 0.15, 0.35, 0.20, 0.20),
  w_pq = c(0.10, 0.10, 0.10, 0.10, 0.10, 0.10, 0.05, 0.10, 0.20, 0.10, 0.10)
)

```

6. Performance Measures

The following measures will be calculated for each scenario.

6.1 Bias

For SI mean:

$$Bias(\hat{\mu}) = \frac{1}{R} \sum_{r=1}^R [\hat{\mu}_r - \mu],$$

where R is the number of simulation replicates.

For SI SD:

$$Bias(\hat{\sigma}) = \frac{1}{R} \sum_{r=1}^R [\hat{\sigma}_r - \sigma].$$

Relative bias may also be reported:

$$Relative\ Bias = \frac{Bias}{True\ Value}.$$

6.2 Root mean squared error

For SI mean:

$$RMSE(\hat{\mu}) = \sqrt{\frac{1}{R} \sum_{r=1}^R [\hat{\mu}_r - \mu]^2}.$$

The same definition is used for the SI SD.

6.3 Empirical standard deviation

The empirical standard deviation of estimates across simulation replicates will be reported for both the SI mean and SI SD:

$$SD_{emp}(\hat{\mu}), \quad SD_{emp}(\hat{\sigma}).$$

This measures variability of the estimator across repeated simulated datasets.

6.4 Convergence rate

The convergence rate is the proportion of simulation replicates for which the fitted EM algorithm reports convergence:

$$Convergence\ rate = \frac{\#\{\text{converged fits}\}}{R}.$$

This is retained as a basic diagnostic because the lognormal extension is more computationally demanding than the original Normal version.